A two term tail certificate for local prime gamma positivity

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Abstract

This consolidated revision gives computer-assisted local certificates for a signed prime–gamma form on its entire real-even minimum form domain. The established ordinary L^2 lower bound is 10^{-20} for support radius $B \le 3/4$; the new $B \le 4/5$ endpoint certificate gives 10^{-22} and is identified separately below. Both constructions retain the complete gamma energy, full pole and every active prime power $2, 3, 4$. A single finite compression of the actual bounded high defect is combined with a signed two-vector low-coupling tail and an N^{-5} residual cost. The extension uses a sharper source-independent diagonal estimate and an exact rank-two displacement identity to generate all off-diagonal source entries from the actual full first row, with separately evaluated diagonals. Old and new numerical datasets are not mixed. The analytic reductions, source interval arithmetic and independent exact-rational replay have distinct trust boundaries. No all-radius positivity, Riemann hypothesis proof, historical priority, human peer review or proof-assistant formalization is claimed.

Verification scope. Both endpoint certificates passed source interval gates and separate exact-rational replays. The complete analytic chains were reviewed. The replays inherit the initial source special-function enclosures and do not establish them independently. These are AI-assisted computational and analytic reviews, not independent human peer review or proof-assistant verification. The previous v1.0 packet is preserved unchanged.

1 The form and the claim

All norms in this paper are ordinary unweighted L^2 norms. Put

$$\rho(s) = \frac{e^{-s/2}}{1 - e^{-2s}}, \qquad a_n = \frac{\Lambda(n)}{\sqrt{n}}, \qquad c_\Gamma = \log \pi + \gamma_E + \frac{\pi}{2} + 3 \log 2.$$

For real even smooth functions, extended by zero outside their support, define

$$t_\Gamma(u) = \int_0^\infty \rho(s) \|u(\cdot + s) - u\|_2^2 \, \mathrm{d}s, \tag{1}$$

$$p(u) = 2 \sum_{n \ge 2} a_n \int_{\mathbb{R}} u(x) u(x + \log n) \, \mathrm{d}x,$$

$$q_B(u) = t_\Gamma(u) - c_\Gamma \|u\|_2^2 - p(u) + 2 \left| \int_{\mathbb{R}} \cosh(x/2) u(x) \, \mathrm{d}x \right|^2. \tag{2}$$

The subscript means restriction by support, not a change in the full-line expression. Let $\mathcal{D}_B$ be the closure of real even $C_c^\infty((-B, B))$ in the norm $(t_\Gamma + \|\cdot\|_2^2)^{1/2}$, with zero extension. On this interval the prime correlations with $\log n \ge 2B$ vanish almost everywhere; the remaining translation forms and pole form are bounded on L^2. Thus q_B is well defined and continuous for form convergence on $\mathcal{D}_B$.

Theorem 1 (Local certificate)**.** *The analytic reduction and finite certificate specified below give*

$$q_B(u) \geq c(B)\|u\|_2^2 \quad (u \in \mathcal{D}_B), \qquad c(B) = \begin{cases} 10^{-20}, & 0 < B \leq 3/4, \\ 10^{-22}, & 3/4 < B \leq 4/5. \end{cases}$$

The finite arithmetic components are identified by the hashes and exact rational evidence in the accompanying packet. Its trust boundary is stated in Section 7.

The finite proof concerns all vectors in the stated domain, not just trial functions. Its small constant is a conservative lower bound, not an approximation of the least eigenvalue. It is neither a theta-weighted spectral gap nor a bound that can be added to constants from another form. Increasing B changes the active prime graph and requires a new certificate.

2 The exact operator on a half interval

Fix $B = 3/4$ until the final support inclusion. Set

$$s = \log 2, \quad t = \log 3, \quad a = s/\sqrt{2}, \quad b = s/2, \quad c = t/\sqrt{3}, \quad \sigma = a + b + c.$$

The weight of 4 is b, not $\log 4/2$. The positive expansion $\log((1+x)/(1-x)) = 2\sum_{j\geq 0} x^{2j+1}/(2j+1)$, with geometric tail bounds, gives

$$56/81 < s < 7/10, \qquad 1 < t < 11/10, \qquad \log(5/3) > 1/2.$$

Consequently $s < B < t < 2s < 2B < \log 5$, so precisely $2, 3, 4$ contribute to p. The terms $n \geq 5$ vanish by support, not by numerical truncation.

Use the unitary map $f(x) = \sqrt{2}u(x)$ on $(0, B)$. A prime shift $0 < r < B$ of weight a_r acts by

$$\begin{aligned}(P_{r,a_r} f)(x) = a_r[&\mathbf{1}_{(0,B-r)}(x) f(x+r) + \mathbf{1}_{(r,B)}(x) f(x-r) \\ &+ \mathbf{1}_{(0,r)}(x) f(r-x)].\end{aligned} \tag{3}$$

For $B \leq r < 2B$ only $a_r \mathbf{1}_{(r-B,B)}(x) f(r-x)$ remains. These bounded self-adjoint operators reproduce the full-line correlations exactly. In particular the translation edges have both orientations; the reflection does not receive an extra factor two. Write $P = P_{s,a} + P_{t,c} + P_{2s,b}$.

Set

$$\omega = \pi/B, \quad k_n = \omega n, \quad b_n = \sqrt{(2 - \delta_{n0})/B}, \quad \epsilon_n = (-1)^n, \quad e_n(x) = b_n \cos(k_n x).$$

These form an orthonormal basis of $L^2(0, B)$. With $\alpha_j = 2j + 1/2$, define

$$d_\Gamma(k) = \sum_{j\geq 0} \frac{2k^2}{\alpha_j(\alpha_j^2 + k^2)},$$

$$D_n = d_\Gamma(k_n) - c_\Gamma = \Re\psi(1/4 + ik_n/2) - \log\pi, \tag{4}$$

$$\mathcal{B}_{nm} = b_n b_m \epsilon_n \epsilon_m \sum_{j\geq 0} \frac{\alpha_j^2(1 - e^{-2\alpha_j B})}{(\alpha_j^2 + k_n^2)(\alpha_j^2 + k_m^2)}, \tag{5}$$

$$c_n = \frac{b_n \epsilon_n \sinh(B/2)}{2(1/4 + k_n^2)}, \qquad p_n = 2c_n.$$

The symbol identity follows from the partial-fraction series for the digamma function and $-\psi(1/4) = \gamma_E + \pi/2 + 3\log 2$.

For the Neumann Laplacian A_N on $(0, B)$, the resolvent kernel is $\cosh(\alpha \min(x, y)) \cosh(\alpha(B - \max(x, y)))/(\alpha \sinh(\alpha B))$. Expanding its hyperbolic factors verifies

$$\begin{aligned}e^{-\alpha|x-y|} + e^{-\alpha(x+y)} = &\, 2\alpha(A_N + \alpha^2)^{-1}(x, y) \\ &- 2(\coth(\alpha B) - 1)\cosh(\alpha x)\cosh(\alpha y).\end{aligned} \tag{6}$$

Expansion of ρ into $\sum_j e^{-\alpha_j s}$ therefore gives the exact positive-series decomposition

$$t_\Gamma(u) = \sum_n d_\Gamma(k_n)|f_n|^2 + \sum_{j\ge 0} 2(\coth(\alpha_j B) - 1)|\langle f, \cosh(\alpha_j x)\rangle|^2.$$

Finite label sums and monotone convergence justify it; integration of each cosine gives (5). The full pole is $4|\langle f, \cosh(x/2)\rangle|^2$, hence pp^*.

For every active shift r, let $\ell = 2B - r$ and $\operatorname{sinc}(0) = 1$. Direct product integration gives

$$\begin{aligned} I_{nm}(r) &= \int_{r-B}^{B} \cos(k_n x)\cos(k_m(x-r))\,\mathrm{d}x \\ &= \frac{\ell}{2}\left[\operatorname{sinc}\left(\frac{(k_n-k_m)\ell}{2}\right)\cos\left(\frac{(k_n+k_m)r}{2}\right)\right. \\ &\qquad \left. + \operatorname{sinc}\left(\frac{(k_n+k_m)\ell}{2}\right)\cos\left(\frac{(k_n-k_m)r}{2}\right)\right]. \end{aligned} \tag{7}$$

For $n > m$ integration of the endpoint Wronskian equivalently gives

$$I_{nm}(r) = -\frac{k_n\epsilon_m \sin(k_n(r-B)) + k_m\epsilon_n \sin(k_m(B-r))}{k_n^2 - k_m^2}.$$

The negative endpoint $s - B$ must retain its sign. Formula (7) is also valid on the diagonal and at the zero mode. The complete form matrix is

$$Q_{nm} = D_n\delta_{nm} + \mathcal{B}_{nm} + p_n p_m - b_n b_m \sum_{r\in\{s,t,2s\}} a_r I_{nm}(r). \tag{8}$$

3 The prime defect and the high comparison

Put $d = B - s$ and partition $(0, B)$ into the center $C = (0, d)$, leaves $L_- = (s - d, s)$ and $L_+ = (s, s + d)$, and middle $J = (d, s - d)$. Write $L = L_- \cup L_+$ and $S = (t - B, B)$. The prime 2 joins $y \in C$ to $s - y, s + y$; prime 4 joins those leaves. Prime 3 maps each leaf to the middle because

$$t - L = (t - B, t - 2s + B) \subset J.$$

Indeed $2s > t$, $t + s > 2B$, and $t + 2B < 4s$ follow from the rational logarithm bounds above.

Define

$$R = \frac{b + c + \sqrt{(b + c)^2 + 8a^2}}{2}, \quad \eta = 2a/R, \quad w = \eta\mathbf{1}_C + \mathbf{1}_{J\cup L}, \quad E = RI - P.$$

The row ratios $(Pw)/w$ equal R on $C \cup L$ and $a + c\mathbf{1}_S$ on J. Thus the weighted edge identity is

$$\begin{aligned} \langle f, Ef\rangle &= a\eta \int_0^d \left(|f(y)/\eta - f(s-y)|^2 + |f(y)/\eta - f(s+y)|^2\right)\,\mathrm{d}y \\ &\quad + b\int_0^d |f(s-y) - f(s+y)|^2\,\mathrm{d}y + \frac{a}{2}\int_J |f(x) - f(s-x)|^2\,\mathrm{d}x \\ &\quad + \frac{c}{2}\int_S |f(x) - f(t-x)|^2\,\mathrm{d}x + \int_J (R - a - c\mathbf{1}_S)|f(x)|^2\,\mathrm{d}x. \end{aligned} \tag{9}$$

Expanding each square proves the identity. Each unoriented edge contributes once; the two reflection integrals have a factor $1/2$. This is an identity of bounded L^2 forms, so it does not require multiplication by w to preserve the gamma domain.

The elementary bounds $a > 7/15$, $a < 1/2$, $c > 4/7$, $c < 121/190$, $b > 1/3$ imply

$$5/4 < R < 27/20, \quad 0 < \eta < 1, \quad \delta := R - a - c > 43/380.$$

For the upper bound, $b + c < 75/76$ and $2a^2 < 49/100$, and substitution of $27/20$ into $X^2 - (75/76)X - 49/100$ gives $1/3800 > 0$. Substitution of $5/4$, using $b > 1/3$, $c > 4/7$ and $2a^2 = s^2 > 4/9$, gives $-13/1008 < 0$. Finally $R > 2a$ follows from $b + c > a$. Weighted Schur comparison on the edges gives $\|P\| \le R$.

Let $e = \langle f, Ef\rangle$ and C_f, L_f, J_f denote squared norms on the three regions. Retaining only center-to-leaf and leaf-to-middle edges in (9) gives

$$J_f \le e/\delta, \qquad L_f \le 2e/c + 2J_f, \qquad C_f \le (\eta/a)e + \eta^2 L_f.$$

The last inequality follows by averaging the two center-to-leaf differences before applying the squared triangle inequality. Hence

$$\|f\|^2 \le \left[\frac{\eta}{a} + \frac{2(1+\eta^2)}{c} + \frac{3+2\eta^2}{\delta}\right] e < \left(\frac{15}{7} + 7 + \frac{1900}{43}\right) e < 54e.$$

We have proved $I/54 \preceq E \preceq 2RI$. This is a defect bound, not positivity of Q by itself.

Split the cosine modes at $M = 32$. On the high space set

$$K_n = D_n - R, \qquad H = E_H + \mathcal{B}_H + p_H p_H^*, \qquad T = K + H.$$

The analytic estimates proved in Appendix A give

$$K_n \ge \kappa := 8/5 \quad (n \ge 32), \qquad K_n \ge \kappa_N := 39/5 \quad (n \ge N := 16384), \qquad I/54 \preceq H \preceq 43I/10. \tag{10}$$

All finite entries of Q still use the signed primes, not R in their place.

4 Domain admission and a single high resource

A zero-extended cosine is in $\mathcal{D}_B$. In fact cut off strips of width ε at the endpoints. The error has bounded amplitude, support $O(\varepsilon)$ and uniformly bounded variation, whence

$$\|w_\varepsilon(\cdot + s) - w_\varepsilon\|_2^2 \le C \min(s, \varepsilon) \quad (0 < s < 1).$$

Since $\rho(s) = O(1/s)$ there and is integrable for $s \ge 1$, its squared form norm is $O(\varepsilon(1 + |\log \varepsilon|))$. The cutoff functions are smooth and supported in the open interval. The gamma operator image of a fixed cosine has at most logarithmic endpoint growth, which is square integrable; the bounded prime and pole terms add no obstruction.

The complete boundary is bounded by Appendix A. Consequently the positive diagonal gamma form plus a bounded term has precisely its diagonal form domain. Cosine partial sums converge there, and the cutoff argument places them in the minimum domain. Conversely, smooth compactly supported even tests have rapidly decreasing cosine coefficients and converge in the logarithmically weighted diagonal norm; the bounded boundary, prime and pole terms preserve that convergence. The decomposition for smooth functions and closure therefore identifies this diagonal domain with $\mathcal{D}_B$. Thus the high operator T has domain $\mathrm{Dom}(K)$ and form domain $\mathrm{Dom}(K^{1/2})$. The high-low map $\mathcal{C}$, with rows $(Q_{nm})_{m<32}$, is bounded because the complete row bound below is $O(1/n)$. Low projections and Schur square completion are legitimate on the original minimum domain.

Let Φ insert the high modes $32, \dots, 63$, and let P_N project onto $32 \le n < N$. Define

$$\mathsf{B} = \Phi^* H \Phi, \qquad F = P_N H \Phi, \qquad E_\infty = (I - P_N) H \Phi.$$

Suppose $E_\infty^* E_\infty \preceq \Theta$. Since $P_N \Phi = \Phi$, we have $\Phi^* E_\infty = 0$. For h and a coefficient vector x, set $g = h - \Phi x$. Expansion gives

$$2\Re\langle h, Fx\rangle - x^*\mathsf{B}x = \langle h, Hh\rangle - \langle g, Hg\rangle - 2\Re\langle g, E_\infty x\rangle \le \langle h, Hh\rangle + 54 x^*\Theta x.$$

Taking the supremum over x after subtraction of the last term proves

$$H \succeq F\mathsf{S}^{-1}F^*, \qquad \mathsf{S} = \mathsf{B} + 54\Theta. \tag{11}$$

There is no separate resource added to this inequality and no extra factor two in S.

The finite sums

$$G = F^*K^{-1}F, \qquad Z = F^*K^{-1}\mathcal{C}$$

only involve $32 \le n < N$. Explicitly, their feature entries are $F_{nj} = R\delta_{nj} + \mathcal{B}_{nj} + p_n p_j - P_{nj}$ for $32 \le j \le 63$, rather than entries of Q with the unbounded diagonal retained. Since $\mathsf{S} + G \succeq I/54$, inverse order and the finite-rank inverse formula imply

$$\mathcal{C}^*T^{-1}\mathcal{C} \preceq \mathcal{C}_0 - Z^*(\mathsf{S} + G)^{-1}Z, \tag{12}$$

for any complete majorant $\mathcal{C}_0 \succeq \mathcal{C}^*K^{-1}\mathcal{C}$. This follows by applying

$$(K + F\mathsf{S}^{-1}F^*)^{-1} = K^{-1} - K^{-1}F(\mathsf{S} + G)^{-1}F^*K^{-1}.$$

Factoring through the bounded $K^{-1/2}$ justifies it despite unbounded K. The selected sufficient matrix is

$$\mathsf{F} = A - \mathcal{C}_0 + Z^*(\mathsf{S} + G)^{-1}Z, \qquad A = (Q_{nm})_{n,m<32}. \tag{13}$$

No other high-feature correction is added.

5 Complete row bounds and the two term tail

Write $v_m = b_m\epsilon_m$, $h_0 = 0$, $h_m = 1 + \pi k_m/4$ for $m > 0$, and

$$U_m = \sum_{j\ge 0}(\alpha_j^2 + k_m^2)^{-1}, \quad E_0 = \frac{e^{-B}}{1 - e^{-4B}}, \quad \delta_m = 1 - (m/N)^2.$$

Decreasing-sum comparison gives $U_n \le 1/(k_n^2 + 1/4) + \pi/(4k_n)$ for $n > 0$. For $n \ge N > m$, set

$$\beta_n = b_n\left[\epsilon_n U_n + \sum_r a_r \sin(k_n(r - B))/k_n\right].$$

The complete off-diagonal row (8) has a first-order expansion

$$Q_{nm} = \beta_n v_m + e_{nm}, \qquad |e_{nm}| \le b_n C_m/k_n^2,$$

where

$$C_m = b_m\left[\frac{h_m}{\delta_m} + E_0 + \frac{\sigma(k_m + k_m^2/k_N)}{\delta_m}\right] + 2\sinh(B/2)|c_m|. \tag{14}$$

Indeed the nonexponential boundary sum equals $U_n + k_m^2(U_n - U_m)/(k_n^2 - k_m^2)$; its remainder uses $k_m^2 U_m \le h_m$. The whole exponential subtraction is bounded by E_0/k_n^2, the prime remainder by $\sigma(k_m + k_m^2/k_N)/(\delta_m k_n^2)$ before normalizations, and the full pole by $2b_n\sinh(B/2)|c_m|/k_n^2$. No gamma label is discarded.

For trial indices $J_t = \{32, \dots, 63\}$ the omitted high rows of H equal those of Q. The squared two-term inequality and integral tests therefore give

$$\Theta = L_H v_t v_t^* + E_H I_{32}, \tag{15}$$
$$L_H = \frac{4(\pi/4 + \sigma + 1/k_N)^2}{B\omega^2(N-1)}, \qquad E_H = \frac{4\sum_{m\in J_t} C_m^2}{3B\omega^4(N-1)^3}.$$

This is an unweighted raw feature-tail Gram; it has no κ_N denominator.

For the low indices $m < 32$ we use a more accurate expansion. Let $s_h = \sinh(B/2)$ and define

$$\begin{aligned} A_m^{\exp} &= \sum_{j\ge 0} \frac{\alpha_j^2 e^{-2\alpha_j B}}{\alpha_j^2 + k_m^2}, \\ w_m &= -v_m(k_m^2 U_m + A_m^{\exp}) + 2 s_h c_m + b_m \sum_r a_r k_m \sin(k_m(B-r)), \\ \gamma_n &= b_n \epsilon_n / k_n^2. \end{aligned} \tag{16}$$

Then $Q_{nm} = \beta_n v_m + \gamma_n w_m + r_{nm}$, with

$$\begin{aligned} |r_{nm}| &\le b_n D_m / k_n^3, \\ D_m &= b_m \left\{ \frac{k_m^2(\pi/4+\sigma)}{\delta_m} + \frac{k_m^2(1+h_m)+\sigma k_m^3}{\delta_m k_N} + \frac{E_2}{k_N} \right\} + \frac{s_h |c_m|}{2k_N}, \\ E_2 &= \sum_{j\ge 0} \alpha_j^2 e^{-2\alpha_j B}. \end{aligned} \tag{17}$$

To check the complete remainder, the nonexponential part after subtraction is $k_m^2 U_n/(k_n^2 - k_m^2) - k_m^4 U_m/[k_n^2(k_n^2 - k_m^2)]$. The remaining exponential subtraction has absolute value at most E_2/k_n^4. The pole remainder is at most $b_n s_h |c_m|/(2k_n^4)$. The two prime remainders are at most $b_n b_m \sigma k_m^2/(\delta_m k_n^3)$ and $b_n b_m \sigma k_m^3/(\delta_m k_n^4)$. These five bounds prove (17) with all signs and channels retained before taking absolute values.

Crucially, the second vector can be recovered from the *original full low matrix*:

$$w_m = v_m(4s_h^2 - E_0) - \frac{k_m^2}{b_0} A_{0m}. \tag{18}$$

For $m > 0$, put $E_m = \sum_j e^{-2\alpha_j B}/(\alpha_j^2 + k_m^2)$. The zero row is $A_{0m} = b_0 v_m(U_m - E_m) + 4c_0 c_m - P_{0m}$, where $P_{0m} = b_0 b_m \sum_r a_r \sin(k_m(B-r))/k_m$. Since $A_m^{\exp} = E_0 - k_m^2 E_m$ and $c_0 = 2b_0 s_h$, substitution cancels the prime terms and combines the pole terms to $4v_m s_h^2$. For $m = 0$, $k_0 = 0$ and $A_0^{\exp} = E_0$ give the same result. The diagonal D_0 is multiplied by zero. Replacing A by a Schur matrix would be incorrect.

For $m \le 31$ use $b_m < 5/3$, $\delta_m^{-1} < 1001/1000$, $k_m \le 21m/5$, $k_N > 65536$, $\pi/4 + \sigma < 16/7$, $s_h < 1/2$, and $|c_m| \le 2 b_m s_h$. With $q = e^{-3} < 1/20$,

$$E_2 < 4\frac{420}{6859} + 2\frac{20}{361} + \frac{1}{4}\frac{20}{19} = \frac{4245}{6859} < 1.$$

The first coefficient of m^2 in (17) is at most $42042/625 < 68$, the second is at most $110213103/819200000 < 1$ (use $m^3 \le 31m^2$), and the constant is at most $25/786432 < 1$. Thus $D_m < 69m^2 + 1 < 70(m+1)^2$.

Apply $|a+b+c|^2 \le 2|a|^2 + 4|b|^2 + 4|c|^2$, and weighted Cauchy–Schwarz to the remainder with weights $(m+1)^4$. Since $\omega > 4$ and $b_n^2 = 2/B$ for $n \ge N$, integral comparison of the n^{-2}, n^{-4}, n^{-6} sums gives the complete inverse-cost tail

$$\sum_{n\ge N} \frac{Q_{n,L}^* Q_{n,L}}{K_n} \preceq L_0 v v^* + \tau_w w w^* + \tau_r \operatorname{diag}_{m<32}(m+1)^4, \tag{19}$$

$$\begin{aligned} L_0 &= L_H/\kappa_N, \\ \tau_w &= \frac{8}{3B\kappa_N 4^4 (N-1)^3} = \frac{5}{12347453440914696}, \\ \tau_r &= \frac{8\cdot 32\cdot 70^2}{5B\kappa_N 4^6 (N-1)^5} = \frac{1225}{13808707107682529275073 1}. \end{aligned}$$

Here $Q_{n,L}$ is the full low row, not its expansion remainder. The two signed vectors are retained, but their mutual cross covariance is bounded by a fixed Young inequality, not evaluated exactly. The order N^{-5} refers to the residual cost; the leading two tails remain of orders N^{-1} and N^{-3}.

Use (19) plus the exact finite sum $\sum_{n=32}^{N-1} Q_{n,L}^* Q_{n,L}/K_n$ as $\mathcal{C}_0$ in (13). The original stored comparison used the valid scalar tail $L_0 vv^* + s_0 I$, where $s_0 = 4\sum_{m<32} C_m^2/[3B\kappa_N \omega^4 (N-1)^3]$, obtained from the first-order row bound by the squared two-term inequality. Its replacement is exactly

$$\mathsf{F}_{\text{two}} = \mathsf{F}_{\text{old}} + s_0 I - \tau_w w w^* - \tau_r \operatorname{diag}(m+1)^4. \tag{20}$$

The raw feature bound Θ, S, G, Z, and the Woodbury correction are unchanged. There is no general Loewner comparison to the old candidate, and no assertion that its earlier failure made the actual form negative.

6 Finite arithmetic and the full norm conversion

The fixed parameters are $B = 3/4$, $M = 32$, $N = 16384$, trial modes 32 through 63, and 256-bit real and complex ball arithmetic using python-flint 0.9.0. The original producer stores the full low matrix A, finite inverse-cost sums, the resource matrices, their correction, and the old sufficient matrix. Although the old sufficient matrix did not pass positivity, its enclosing arrays remain valid mathematical input. The second-stage calculation imports their exact rational endpoints, reconstructs (18), and applies (20). No digamma values, prime phases, or source rows are recomputed at this stage; elementary enclosures of π, square roots, $\sinh(3/8)$, $e^{-3/4}$ and e^{-3} are evaluated afresh. Interval imports are containing unions, not midpoint substitutions.

The complete nonexponential sums in the source are evaluated using

$$U(0) = \psi_1(1/4)/4, \qquad U(k) = \Im\psi(1/4 + ik/2)/(2k), \quad k > 0,$$

$$\sum_j \frac{\alpha_j^2}{(\alpha_j^2 + k_n^2)(\alpha_j^2 + k_m^2)} = \frac{k_n^2 U_n - k_m^2 U_m}{k_n^2 - k_m^2}, \qquad n \neq m.$$

On the positive diagonal the corresponding formula is

$$\frac{\Im\psi(1/4 + ik/2)}{4k} + \frac{\Re\psi_1(1/4 + ik/2)}{8};$$

the zero diagonal is $U(0)$. Thus no coincident-frequency divided difference is used. Only the exponential subtraction is cut at label $j = 32$. Its omitted positive inner sum is enclosed by

$$[0, \Delta_{32}], \qquad \Delta_{32} = \frac{e^{-2\alpha_{32}B}}{\alpha_{32}^2(1 - e^{-4B})} = \frac{4e^{-387/4}}{16641(1 - e^{-3})}.$$

This follows by bounding each denominator by α_j^4, then summing a geometric series. The subtraction is enclosed with its negative sign before the signed normalizations; a symmetric outward enclosure is also valid. This treats the entire gamma label tail.

Every matrix stored for the finite target is symmetrized by containing unions. A fixed-index interval LDL^* factorization of F_{two} has all 32 pivot lower endpoints strictly positive. The least displayed pivot is approximately 5.18514×10^{-6}, but this is *not* the eigenvalue lower bound. An inverse-factor norm conversion is essential because the unit triangular factor is ill-conditioned.

A separate standard-library rational calculation reconstructs a containing interval matrix $\mathcal{I}$ directly from the original arrays and the saved elementary enclosures, and takes its hull with the second-stage matrix. Let L_0, D_0 be rational midpoints of the saved triangular factor and diagonal pivots. Let $\mathcal{I}_{ij} = [\ell_{ij}, u_{ij}]$ and define

$$\varepsilon = \max_i \sum_j \left(\left| \frac{\ell_{ij} + u_{ij}}{2} - (L_0 D_0 L_0^*)_{ij} \right| + \frac{u_{ij} - \ell_{ij}}{2} \right),$$

$$U_* = \|L_0^{-1}\|_F^2, \qquad \mu = \frac{\min_i (D_0)_{ii}}{U_*} - \varepsilon. \tag{21}$$

The residual is symmetric, so its operator norm is at most its maximum absolute row sum. Moreover $L_0 D_0 L_0^* \succeq (\min D_0)/U_* I$. These facts prove $\mathsf{F}_{\mathrm{two}} \succeq \mu I$. The inverse is computed by exact triangular recurrence; all statements in (21) are rational operations, not rounded decimal comparisons.

For a complete cost trace use

$$\tau \geq \operatorname{tr}(\mathcal{C}_{0,\mathrm{old}}) + \tau_w \|w\|^2 + \tau_r \sum_{j=1}^{32} j^4.$$

Dropping the favorable $-32 s_0$ is conservative. The rational checker enlarges the saved trace upper bound if required by its independently reconstructed w intervals. Approximate values are listed only for readability:

Quantity	Rounded display
Positive pivots	32
Residual row norm ε	2.29635×10^{-30}
Inverse triangular Frobenius norm squared U_*	3.46080×10^{14}
Independent low margin μ	$1.49824946936 \times 10^{-20}$
Complete coupling trace upper bound τ	0.288227
Full ordinary L^2 lower bound g	$1.10142502550 \times 10^{-20}$

The exact rational strings in the evidence, not this rounded table, certify the strict comparison to 10^{-20}.

For completeness, write $f = z + h$ and $t_0 = \|T^{1/2}(h + T^{-1}\mathcal{C}z)\|^2$. The exact Schur square and (12) give $q_B(u) \geq \mu\|z\|^2 + t_0$. Also

$$\|T^{-1}\mathcal{C}\|^2 \leq \kappa^{-1}\|\mathcal{C}^* T^{-1}\mathcal{C}\| \leq \tau/\kappa.$$

The second inequality uses $T^{-1} \preceq K^{-1}$ and the complete cost envelope. Hence

$$\|u\|_2^2 \leq (1 + 2\tau/\kappa)\|z\|^2 + (2/\kappa)t_0, \qquad g = \min\left\{\frac{\mu\kappa}{\kappa + 2\tau}, \frac{\kappa}{2}\right\} > 10^{-20}.$$

This proves the claimed bound at $B = 3/4$. All preceding form inequalities extend from the core by the domain identification. If $B' \leq 3/4$, then $\mathcal{D}_{B'} \subset \mathcal{D}_{3/4}$; the same zero-extended function has exactly the same full-line form and norm. Thus the bound extends to every smaller positive radius.

7 Reproducibility and scope

The 3/4 packet separates three different checks; the fresh 4/5 reconstruction is detailed in Section 8. The source producer encloses transcendental matrix entries and complete finite sums. The saved-data interval gate replaces one analytic tail envelope and performs a finite factorization. The independent rational checker verifies the saved matrix reconstruction, residual, inverse-factor bound and final norm conversion without new special-function evaluations. It does not independently establish the initial transcendental enclosures. Hashes establish identity, not mathematical truth.

The proof therefore relies on the analytic reductions given here, their implementation by the preserved source, and the correctness of Python rational arithmetic and python-flint's Arb/Acb interval operations. Neither positive sampled Rayleigh quotients nor finite-matrix positivity without tails would suffice. Every infinite component used here is covered by (15), (19), or Δ_{32}; absent prime powers vanish exactly by support.

The original sources retain their preparatory headers and explicit execution guards as provenance. Those headers do not replace the saved outcome. The earlier sufficient comparison failed;

only the two-term replacement is asserted positive. Operational launch receipts and private approval records are excluded from the public packet. The README distinguishes preserved executed sources from the separately tested portable exact-rational replay. That replay was executed successfully in a clean staging directory; it does not regenerate the transcendental source arrays. Compiling the manuscript does not run a numerical producer.

For context, Zhu's preprint [2] claims a compact-window bound at radius 0.8, including both parities. We have not reproduced its stated certificate and do not use that claim as a lemma. The present radius is not presented as a priority record, nor as exceeding all prior claimed radii. The contribution described here is the explicit signed two-vector tail, its recovery from saved low-form data, and the disclosed evidence for this particular local certificate.

AI assistance was used in exploration, derivation, code preparation, review and exposition. Independent computational paths and analytic reviews described here are not independent human peer review. No proof-assistant formalization is claimed. No conclusion about arbitrary support, global 1/2 positivity, or RH follows from these bounded radii.

8 The consolidated extension to four fifths

This section adds a second, freshly computed endpoint certificate at $B = 4/5$; it does not replace any $B = 3/4$ array by a rescaling. The preceding proof and stronger bound 10^{-20} remain valid for $B \le 3/4$. The definitions of the form, minimum domain, half-interval identification and complete matrices are unchanged, with the actual new radius in every frequency and normalization. The new lower bound is 10^{-22}. It is a consolidated revision of the same local-certificate study, not a claim of a second historically novel theorem.

8.1 Uniform geometry and fresh high floors

For $z = (x-1)/(x+1)$ define $L_6(x) = 2\sum_{j=0}^{5} z^{2j+1}/(2j+1)$. The exact series remainder satisfies

$$L_6(x) < \log x < L_6(x) + \frac{2z^{13}}{13(1-z^2)}.$$

For $x = 2$ the two endpoints are 15757912/22733865 and 2458234657/3546482940; for $x = 3$ they are 3897967/3548160 and 25337363/23063040. Also $L_6(5) = 36559324/22733865 > 8/5$. These rational comparisons give $69/100 < s < 7/10$, $109/100 < t < 11/10$ and $\log 5 > 8/5$. Uniformly for $3/4 \le B \le 4/5$,

$$s < B < 3s/2, \quad B < t < 2s < 2B < \log 5, \quad B < (s+t)/2, \quad t + 2B < 4s.$$

Thus exactly $2, 3, 4$ remain active and the center–leaves–middle partition, including $t - L \subset J$, remains valid. The proof of (9) and the subsequent norm estimates therefore proves the same $I/54 \preceq RI - P$, without assuming that a previous numerical result transfers. The constants R, a, b, c, δ, η depend only on the primes, not on B.

A sharper source-independent symbol estimate is useful here:

$$D(k) \ge \log\frac{k}{2\pi} - \frac{2}{81} - \frac{21}{2k^2} \qquad (k > 0). \tag{22}$$

To prove it, retain the first two summands in the harmonic argument of Appendix A. The trapezoidal error now begins at 9/4 and gives

$$C_\Gamma = 4 + 4/5 + 2/9 - \log(9/4) + E, \qquad 0 \le E \le 2/81.$$

Convexity of F_k gives

$$d_\Gamma(k) \ge F_k(1/2) + F_k(5/2) + \tfrac{1}{2}\log(1 + k^2/(9/2)^2) + \tfrac{1}{2}F_k(9/2).$$

The rational terms expand into the constants $4 + 4/5 + 2/9$ minus

$$\frac{1}{k^2 + 1/4} + \frac{5}{k^2 + 25/4} + \frac{9/2}{k^2 + 81/4} \le \frac{21}{2k^2}.$$

The remaining logarithm is nonnegative. Cancellation proves (22); neither a digamma evaluation nor an asymptotic expansion is used.

At $B = 4/5$, $k_{32} = 40\pi > 120$ and $k_N = 20480\pi > 61440$. In addition to $s > 56/81$, the four-term logarithm sum gives $\log 5 > 122492/76545$. Substitution into (22), using $\log 20 = 2s + \log 5$, $\log 10240 = 11s + \log 5$ and $R < 27/20$, yields

$$\begin{aligned} K_{32} &> \frac{78752279}{48988800} = \frac{8}{5} + \frac{370199}{48988800}, \\ K_N &> \frac{100560325801079}{12842119987200} = \frac{39}{5} + \frac{391789900919}{12842119987200}. \end{aligned} \tag{23}$$

Monotonicity in k proves the same floors $\kappa = 8/5$ and $\kappa_N = 39/5$ for all high modes at the new radius, and uniformly on the intervening radius interval. No old-radius scalar source endpoints are needed.

The general boundary norm proof gives, at the new radius,

$$\|\mathcal{B}_H\| \le \pi/2 + \frac{2B}{63\pi^2} < \frac{11}{7} + \frac{8}{2835} = \frac{4463}{2835}.$$

The series bound $\sinh x \le x/(1 - x^2/6)$ for $0 \le x \le 2/5$, obtained from $(2j+1)! \ge 6^j$, gives $\sinh(B/2) \le 30/73 < 1/2$. Since $\omega > 15/4$ and $b_n^2 \le 8/3$,

$$\|p_H\|^2 < \frac{2/3}{3(15/4)^4\, 31^3} = \frac{512}{13573524375}.$$

The sum of the displayed boundary and pole bounds is strictly below $8/5$. Thus $I/54 \preceq H \prec 43I/10$ and the domain and one-resource proofs apply at this actual radius.

8.2 The actual new tail constants

The exact two-term row expansion (17)–(18) is algebraic in the radius and holds at $B = 4/5$. For $m \le 31$, throughout $3/4 \le B \le 4/5$,

$$b_m < 5/3, \quad \delta_m^{-1} < 1001/1000, \quad k_m \le 21m/5, \quad k_N > 61440, \quad E_2 < 1.$$

The last inequality follows because E_2 decreases with B. The three estimates for the coefficients of D_m become

$$\frac{42042}{625} < 68, \qquad \frac{36737701}{256000000} < 1, \qquad \frac{5}{147456} < 1.$$

Consequently the same $D_m < 70(m+1)^2$ holds. With $B = 4/5$ and the lower frequency bound $\omega > 15/4$, the complete new low-cost tail is

$$\begin{aligned} \sum_{n \ge N} \frac{Q_{n,L}^* Q_{n,L}}{K_n} &\preceq L_0 v v^* + \tau_w^{(080)} w w^* + \tau_r^{(080)} \operatorname{diag}(m+1)^4, \\ \tau_w^{(080)} &= \frac{8}{3(4/5)(39/5)(15/4)^4 (N-1)^3} = \frac{512}{1041816384077177475}, \\ \tau_r^{(080)} &= \frac{8 \cdot 32 \cdot 70^2}{5(4/5)(39/5)(15/4)^6 (N-1)^5} = \frac{51380224}{419439478395856826730345 4125}. \end{aligned} \tag{24}$$

Here L_0, v, w, and the finite cost sum are freshly evaluated at $B = 4/5$. The coefficients in (24) are outward rational majorants; no upper frequency bound has been substituted in a denominator.

The raw trial-tail formula (15) is rebuilt with the actual new C_m for $32 \le m \le 63$ and still has no κ_N divisor. The exponential-label remainder is now

$$\Delta_{32}^{(080)} = \frac{4e^{-516/5}}{16641(1-e^{-16/5})}.$$

All other finite source and compression definitions are those of the preceding sections evaluated at 4/5, not transported stored matrices.

8.3 An exact structured source identity

There is an exact simplification of off-diagonal source generation. For all needed indices let $a_j^{(0)} = Q_{0j}$ be the *actual full first row*, and put $d_j = v_j/b_0 = \epsilon_j\sqrt{2-\delta_{j0}}$. For $n \neq m$,

$$Q_{nm} = \frac{n^2 d_m a_n^{(0)} - m^2 d_n a_m^{(0)}}{n^2 - m^2}. \tag{25}$$

It does not determine Q_{nn}; those diagonal entries are computed separately.

To prove it, use the complete w_j of Section 5. Multiplying the full off-diagonal formula by $k_n^2 - k_m^2$ gives

$$(k_n^2 - k_m^2)Q_{nm} = v_n w_m - w_n v_m.$$

The nonexponential boundary gives $v_n v_m(k_n^2 U_n - k_m^2 U_m)$. Its exponential subtraction gives $v_n v_m(A_n^{\exp} - A_m^{\exp})$ by partial fractions. The full pole gives $2\sinh(B/2)(v_n c_m - c_n v_m)$, and the signed prime endpoint gives $v_n b_m k_m \sum_r a_r \sin(k_m(B-r)) - b_n k_n \sum_r a_r \sin(k_n(B-r))v_m$. These are exactly the four terms of the displayed wedge identity. All gamma sums converge absolutely. Now substitute $w_j = (4\sinh^2(B/2) - E_0)v_j - (k_j^2/b_0)a_j^{(0)}$ from (18). The first scalar term cancels, and the scale ω^2 cancels from numerator and denominator, proving (25).

Thus every finite compression has a displacement of rank at most two with respect to $\mathrm{diag}(k_j^2)$. This is a componentwise and finite-compression identity only: $v \notin \ell^2$, so no bounded rank-two infinite-operator commutator is asserted. The identity does not differentiate across zero-extension jumps.

The first-row input is needed for *every* required index, including high n. A stored 32-entry low row alone cannot supply the new rectangular source. With exact d_j and input radii e_j, the error propagated through (25) is at most

$$\frac{n^2|d_m|e_n + m^2|d_n|e_m}{|n^2 - m^2|}.$$

Actual computation uses outward intervals also for $\sqrt{2}$, an exact nonzero integer denominator, and the unchanged diagonal formula. It never substitutes midpoints for source balls. Near-diagonal error amplification is thereby included rather than ignored.

The new producer freshly encloses Q_{0n} for $0 \le n < N$, separately encloses all needed diagonals, and reconstructs off-diagonal entries by (25). It accumulates 128-row blocks in combined column order $(\mathcal{C}, F)$. The product of the transpose with the row-weighted matrix has blocks $\mathcal{C}_{\mathrm{finite}}$, G, and $Z = F^* K^{-1}\mathcal{C}$ in their corresponding positions. This changes only evaluation organization, not the one-resource inequality or the sign of a prime phase. No controlled speed comparison or performance record is claimed.

8.4 The new finite certificate and its boundary

The new source interval gate and separate exact-rational replay both passed. The latter does not trust the producer's printed positive gap as its conclusion. It reconstructs the low off-diagonal entries by (25), uses independent integer-square-root enclosures, reconstructs the complete cost

and resource matrices, and checks the inverse and final Schur margins by residuals. The complete analytic chain has also been reviewed within the stated source-enclosure trust boundary.

For precision, write $M_r = \mathsf{B} + 54\Theta + G$ and let X be the symmetric rational midpoint of a stored inverse enclosure. Exact rational interval products bound $I - M_r X$ by an operator norm bound $r < 1$, using the larger of its maximum absolute row and column sums. If $x_* \geq \|X\|$, the Neumann identity gives

$$\|M_r^{-1} - X\| \leq \varepsilon_{\mathrm{inv}} := x_* r/(1-r), \qquad M_r^{-1} \succeq X - \varepsilon_{\mathrm{inv}} I.$$

Therefore it is enough to verify the lower comparison

$$A - \mathcal{C}_0 + Z^* X Z - \varepsilon_{\mathrm{inv}} z_*^2 I, \qquad z_*^2 \geq \|Z\|_F^2.$$

Here $\|Z\|_F$ denotes the Frobenius norm of Z. All entries are enclosed by exact rational operations, with outward rounding to a fixed 240-bit dyadic grid at specified matrix boundaries. This is enclosure arithmetic, not floating-point rounding. The saved LDL witness is checked against that independently reconstructed lower comparison by the residual method (21). Its triangular inverse is also checked by exact multiplication.

The replay gives the following rounded displays; the proof comparisons use its exact rational strings:

New radius quantity	Rounded display
Inverse residual bound r	1.58408×10^{-44}
Inverse error bound $\varepsilon_{\mathrm{inv}}$	2.22157×10^{-43}
LDL residual bound	1.02528×10^{-24}
Inverse triangular Frobenius norm squared	8.77561×10^{16}
Independent low margin μ	$2.06064782120 \times 10^{-22}$
Complete coupling trace upper bound τ	0.340068
Independent full ordinary L^2 gap	$1.44598358915 \times 10^{-22}$

The exact final comparison exceeds 10^{-22}. The source interval gate's approximately 1.453178×10^{-22} gap is slightly larger; the claim uses the smaller independently reconstructed bound and rounds it down to 10^{-22}. The new sufficient matrix is assembled directly as

$$\mathsf{F}_{080} = A_{080} - \mathcal{C}_{0,080} + Z_{080}^* (\mathsf{S}_{080} + G_{080})^{-1} Z_{080}.$$

No saved 3/4 matrix is an input to this calculation. The original interval gate uses fixed-order factorization with all 32 pivots positive; the exact-rational replay and final norm conversion are reported separately in the evidence packet. The full L^2 conversion is the same proved formula as before, with the new low margin and new complete cost trace. The exact replay gives $g > 10^{-22}$ at $B = 4/5$. The same minimum-domain support inclusion then proves this bound for all $0 < B \leq 4/5$, completing the second part of Theorem 1. The smaller certified number at the larger radius does not weaken the already established 10^{-20} bound on the smaller interval.

A Analytic diagonal and bounded boundary estimates

We supply the auxiliary bounds used in (10), rather than referring to a previous audit.

Lemma 1. *For $k > 0$, $D(k) \geq \log(k/(2\pi)) - 2/25 - 7/(2k^2)$.*

Proof. Put $C_\Gamma = c_\Gamma - \log \pi$. The harmonic identity

$$C_\Gamma = \lim_{L \to \infty} \left(\sum_{j=0}^{L-1} \frac{1}{j + 1/4} - \log L \right)$$

follows from $\sum_{j<L}(j+1/4)^{-1} = 2H_{4L} - H_{2L} + 2\sum_{j<L}((4j+1)^{-1} - (4j+3)^{-1})$ and the arctangent series. The trapezoidal error for $1/x$ on $[5/4, \infty)$ gives

$$E = C_\Gamma - 4 + \log(5/4) - 2/5, \qquad 0 \le E \le 2/25.$$

Indeed the error on a unit interval is $\frac{1}{2}\int_0^1 t(1-t)(1/(x+t))''\,\mathrm{d}t$; summation and $t(1-t) \le 1/4$ bound it by $1/(8x^2)$. For $F_k(a) = 2k^2/[a(a^2+k^2)]$,

$$F_k''(a) = \frac{4k^2(6a^4 + 3a^2k^2 + k^4)}{a^3(a^2+k^2)^3} > 0.$$

The mesh-two trapezoidal inequality, with the first label retained, yields

$$d_\Gamma(k) \ge F_k(1/2) + \tfrac{1}{2}\log(1 + k^2/(5/2)^2) + \tfrac{1}{2}F_k(5/2).$$

Insert the expression for E, cancel constants, discard the nonnegative residual logarithm, and bound the two negative rational remainders by $1/k^2$ and $(5/2)/k^2$. □

Every term in $d_\Gamma(k)$ increases with $k \ge 0$. At $M = 32$, $k_M = 128\pi/3 > 128$ and $\log(k_M/(2\pi)) = 6s - t$. Since $R < 27/20$,

$$K_{32} > 6\frac{56}{81} - \frac{11}{10} - \frac{2}{25} - \frac{27}{20} - \frac{7}{32768} = \frac{8}{5} + \frac{49}{2700} - \frac{7}{32768} > \frac{8}{5}.$$

At $N = 16384$, $\log(k_N/(2\pi)) = 15s - t$, and the same bound gives

$$K_N > \frac{39}{5} + \frac{109}{2700} - \frac{7}{32768} > \frac{39}{5}.$$

Monotonicity proves both floors, without importing numerical floors from another radius.

To bound the whole boundary, for $x, y > 0$ the function $F(a) = a^2/((a^2+x^2)(a^2+y^2))$ is unimodal, with maximum $(x+y)^{-2}$ and integral $\pi/(2(x+y))$. Every superlevel interval contains at most half its length plus one points of the lattice $1/2 + 2\mathbb{Z}$. Layer-cake integration therefore gives

$$\sum_{j\ge 0} F(\alpha_j) \le \frac{\pi}{4(x+y)} + \frac{1}{(x+y)^2}.$$

Conjugating by the parity signs and discarding $1 - e^{-2\alpha_j B} \le 1$ shows, for $n, m \ge M \ge 1$,

$$|\mathcal{B}_{nm}| \le \frac{1}{2(n+m)} + \frac{2B}{\pi^2(n+m)^2}.$$

With weights $m^{-1/2}$, the decreasing integral bound $\sum_{m\ge M}((n+m)\sqrt{m})^{-1} \le \int_0^\infty ((n+x)\sqrt{x})^{-1}\,\mathrm{d}x = \pi/\sqrt{n}$ gives norm at most π for $(n+m)^{-1}$. The row sums of $(n+m)^{-2}$ are at most $(2M-1)^{-1}$. The symmetric weighted Schur inequality follows directly from $2|z_n z_m| \le |z_n|^2 w_m/w_n + |z_m|^2 w_n/w_m$. Thus

$$\|\mathcal{B}_H\| \le \frac{\pi}{2} + \frac{2B}{\pi^2(2M-1)} < \frac{11}{7} + \frac{1}{378} = \frac{85}{54}$$

at $B = 3/4, M = 32$. The finitely many missing low rows are square summable by the row bounds, so the entire boundary is bounded as used in the domain argument.

Since $\sinh(B/2) < 1/2$, $b_n^2 = 8/3$, $k_n > 4n$, and $n \ge 32$,

$$\|p_H\|^2 < \frac{1}{384}\sum_{n\ge 32} n^{-4} \le \frac{1}{1152 \cdot 31^3} < \frac{7}{270} = \frac{8}{5} - \frac{85}{54}.$$

Therefore $\mathcal{B}_H + p_H p_H^* \preceq (8/5)I$ and $H \preceq 2RI + (8/5)I \prec (43/10)I$. Its lower bound $I/54$ was proved by the edge identity.

B Evidence identification

The packet manifest gives SHA-256 hashes of every public input, source and result. The principal immutable inputs are:

Original source producer `f7288fe5eb238d2e1350fad683c0c4290ba9f4c6fdfb5c891e6457e2c834b00e`

Original containing diagnostic `be35a1bfedcbe49e775c4855fe740a846a0d85173a9630c505b60de1f4d0e4b3`

Two-term interval gate source `845d571fd58ea4e4d950b481d8f98ce134ba3f7dc1bdc7b12ca7e7280dbb918a`

Two-term successful interval result `44db810221f4693d4807f9f3c54f3e746b0bb44ead44f5a9f85a9faba04970db`

Portable exact checker source `5b1ccf4508e25be004739de490bebe1f3858f03f852a1c77ec120541b7e8ec5f`

Portable exact replay result `5c3dd4280ee068d09b841448d4fcd7c6d80903cdbd592784ba1b66702eec6621`

These six identifiers concern the $B = 3/4$ dataset, retained unchanged in `repro_075`. The fresh $B = 4/5$ dataset is separately identified as follows:

New structured source producer `3b46a081b272146cc6cc8474416f124726a45453f55e01bb85b0eb1762e192bc`

New source diagnostic `4c67f0c3f37bb8ce3c4f7c6da166050f2a1760c62bd47227c7709fb3a0b2ea05`

New interval success record `19156f6635bb25e853b9a1aecd650672043113f85d279c562e1a090269b0ca91`

New portable exact checker `3c5e006df82fbc985ce9b7ed73e26d62a9679086cc399dcb022224325c9fde13`

New portable exact result `f2d4eef6e3efcd632ee252b2cca56be120e5ce5b3c59cdcc499277bc4ee528e2`

These identify preserved source material and the two successfully executed portable replays. The new replay is in `repro_080`; neither replay loads the other radius's dataset. The portable checkers have no operational receipt dependencies; their distinct source hashes and results are shown above. The project repository is `https://github.com/serabing-hash/theta-weighted-certificate`. This identifies the project, not an asserted upload or a DOI for this packet.

References

[1] The python-flint developers, *python-flint 0.9.0 documentation*, real and complex ball arithmetic interfaces. `https://python-flint.readthedocs.io/en/stable/arb.html`; `https://python-flint.readthedocs.io/en/stable/acb.html`.

[2] Xuefeng Zhu, *Weil positivity in compact windows: a finite reduction, certified two-sided bounds, and a Landau–Widom decay law*, arXiv:2608.24827v2, 2026. Preprint; its computational claim is not used as a proof dependency here. `https://arxiv.org/html/2608.24827v2`.